Operating Systems · 21CSC202J

Unit 1 · Session 6 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 6 : SLO â€“ 1 | Process Life Cycle â€” Fill in the Missing Words

I. Completed Sentences (Process Life Cycle)

• A PROCESS is a program in execution.
• A process passes through states: NEW, READY, RUNNING, WAITING (BLOCKED) and TERMINATED.
• The OS creates a process and assigns it a PROCESS CONTROL BLOCK (PCB) holding its state,
• program counter, registers and memory information.
• The SCHEDULER (dispatcher) picks a ready process and gives it the CPU.
• When a running process requests I/O it moves to the WAITING state.
• After I/O completes, the process returns to the READY queue.
• When a process finishes execution it is TERMINATED and its resources are deallocated.
• CONTEXT SWITCHING saves the state of one process and restores the state of another.

II. Role of the OS

• Creation: fork()/spawn, PCB allocation, memory setup.
• Scheduling: choosing which ready process runs next (short-term scheduler).
• Termination: exit(), releasing memory, closing open files, removing the PCB.